

Job Shop Scheduling Tutorial

You should do this tutorial using Google Collabatory and using [Google OR-Tools](#).

Note you will need to run the following first to install the ortools python package.

```
!pip install ortools
```

Exercises

1. Job Shop Scheduling Problem:

Jobs	Tasks	Sequence on Machines	Process Time on Machines
J_0	t_{01}, t_{02}, t_{03}	M_0, M_1, M_2	3,2,2
J_1	t_{11}, t_{12}, t_{13}	M_0, M_2, M_1	4,3,1
J_2	t_{21}, t_{22}, t_{23}	M_1, M_3, M_2	2,3,2

2. Job Shop Scheduling Problem:

Jobs	Tasks	Sequence on Machines	Process Time on Machines
J_1	$t_{11}, t_{12}, t_{13}, t_{14}$	A, B, C, D	3,2,4,3
J_2	t_{21}, t_{22}, t_{23}	B, A, D	5,3,2
J_3	$t_{31}, t_{32}, t_{33}, t_{34}$	C, D, A, B	2,1,4,4
J_4	t_{41}, t_{42}, t_{43}	D, C, B	4,3,5

3. Job Shop Scheduling Problem:

Jobs	Tasks	Sequence on Machines	Process Time on Machines
J_A	$t_{A1}, t_{A2}, t_{A3}, t_{A4}, t_{A5}$	M_1, M_2, M_3, M_4, M_5	4,2,3,5,2
J_B	$t_{B1}, t_{B2}, t_{B3}, t_{B4}$	M_3, M_4, M_2, M_5	2,1,3,4
J_C	$t_{C1}, t_{C2}, t_{C3}, t_{C4}$	M_1, M_2, M_3, M_4	3,4,2,3

4. N-queens Problem:

Read through the **N-queens problem** on the [Google OR-Tools site](#).

Implement the solution in Google Collabatory and make sure you understand what the model is doing and how it translates to the Python code.

5. Employee Scheduling Problem:

Read through the **Employee Scheduling Problem** on the [Google OR-Tools site](#).

Implement the solution in Google Collabatory and make sure you understand what the model is doing and how it translates to the Python code.